

MORE WORKED OUT EXAMPLES

1. Design a combinational logic circuit with 3 input variables that will produce logic high output when more than one input variables are logic low. [2071 Chaitra]

⇒ Total number of inputs = 3

Number of output = 1

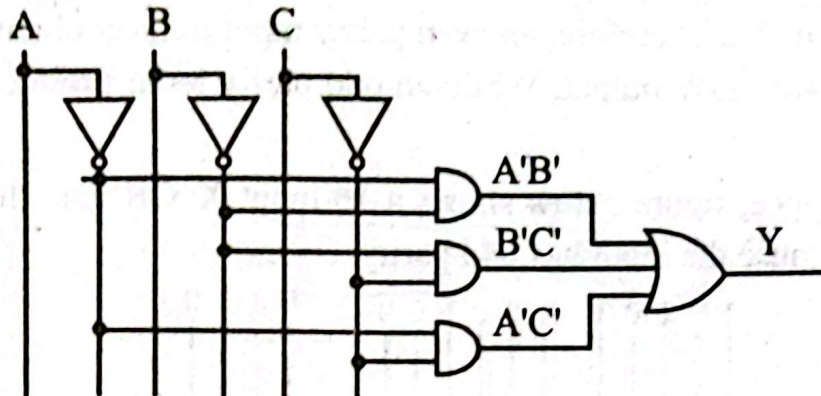
Truth table

A	B	C	Y
0	0	0	1
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	0

Implementing the K-map of above truth table, we get

BC \ A	00	01	11	10
0	1	1	0	1
1	1	0	0	0

$$Y = A'B' + B'C' + A'C'$$



2. Design a 32 to 1 multiplexer using 16 to 1 and 2 to 1 multiplexers.

⇒

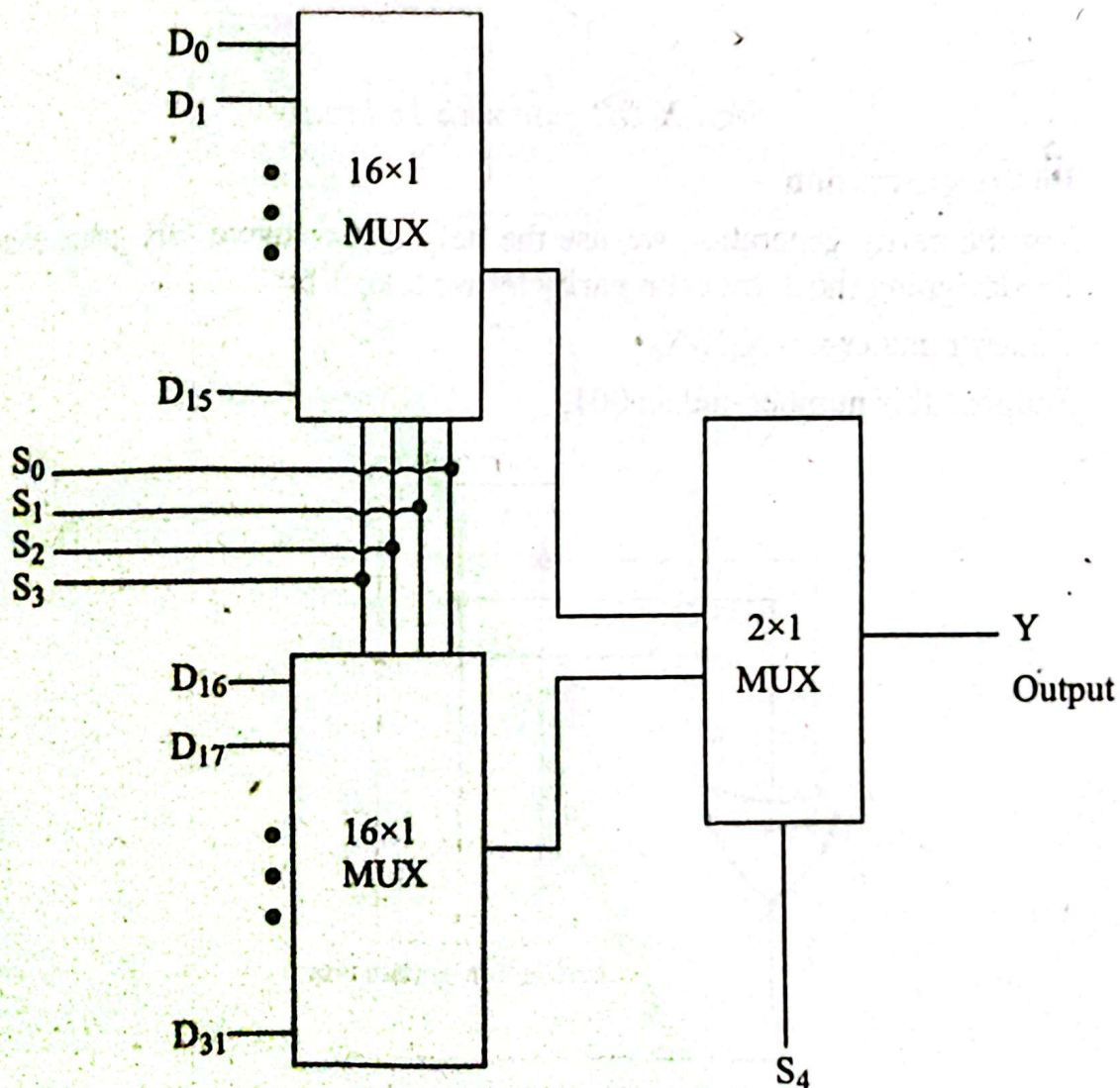


Fig.: Realization of 32x1 MUX by using 16x1 and 2x1 MUXes.

3. Design a 3-bit even parity generator and parity checker.

[2068 Shrawan]

⇒ Even parity means an n-bit input has an even number of 1s. For instance, 110011 has even parity because it contains four 1s.

Parity checker

Exclusive OR (EX-OR) gates ideal for checking the parity of a binary number because they produce an output 1 when the input has an odd number of 1's. Therefore, an even parity input to an exclusive OR gate produces a LOW output. While an odd parity input produces a HIGH output.

For instance, figure below shows a 16-input X-OR gate the output is of 1, because the input has odd parity.

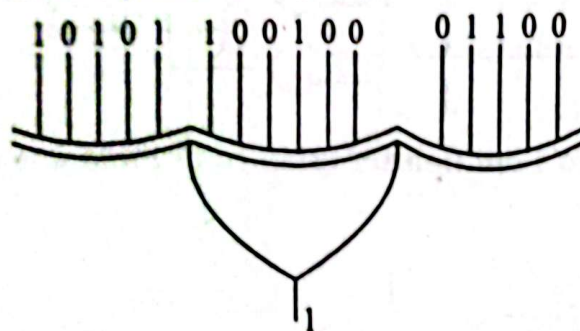


Fig.: X-OR gate with 16 inputs

Parity generation

For the parity generation we use the help of exclusive OR gate. Here for designing the 3-bit even parity let we take 3 bit

Binary number $x = X_2X_1X_0$

Suppose this number equals 001.

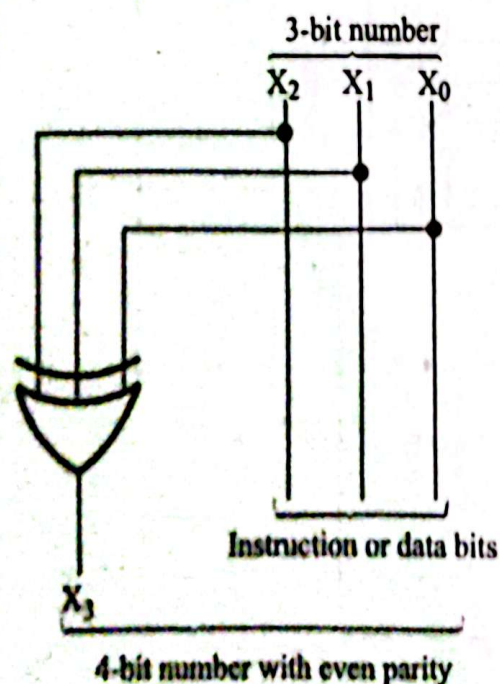


Fig.: Even parity generation

Since the 3 bits (input bits) are 001. Then the number has odd parity, which means the exclusive OR gate produces an output of 1.

The final 4-bit output is 1001. Notice that this has even parity.

4. Simplify $F = \Sigma (0, 2, 4, 6, 8, 9, 13, 14)$ by using K-map. Implement the given circuit using decoder (use only the block diagram of decoder). [2068 Shrawan]

$\Rightarrow F(A, B, C, D) = \Sigma(0, 2, 4, 6, 8, 9, 13, 14)$

Using K-map:

CD \ AB	00	01	11	10
00	1			1
01	1			1
11		1		1
10	1	1		

In SOP form,

$$F(A, B, C, D) = A'D' + B'C'D' + AC'D + BCD'$$

Now, designing the given functional system by using decoder.

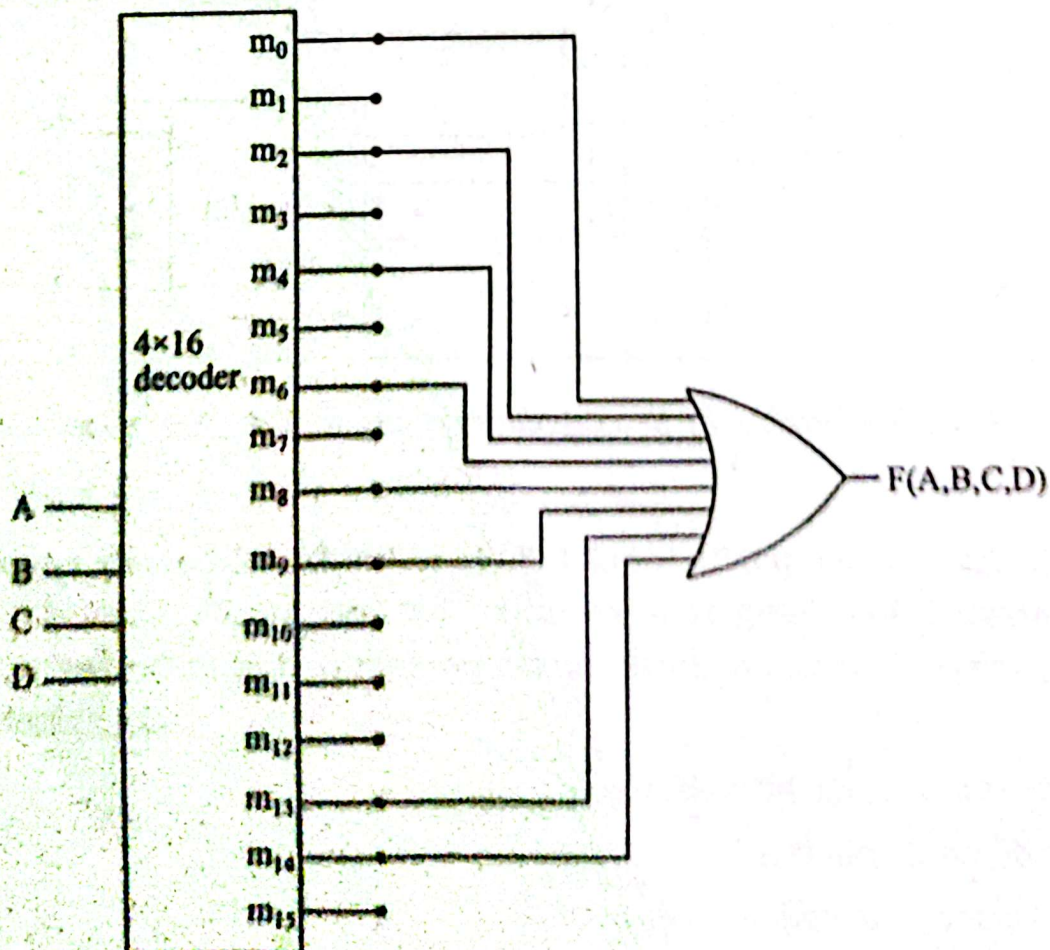


Fig.: 4x16 decoder circuit

5. Construct a 5×32 decoder using 3 to 8 decoders and standard logic gates if necessary. [2067 Ashadh]

⇒

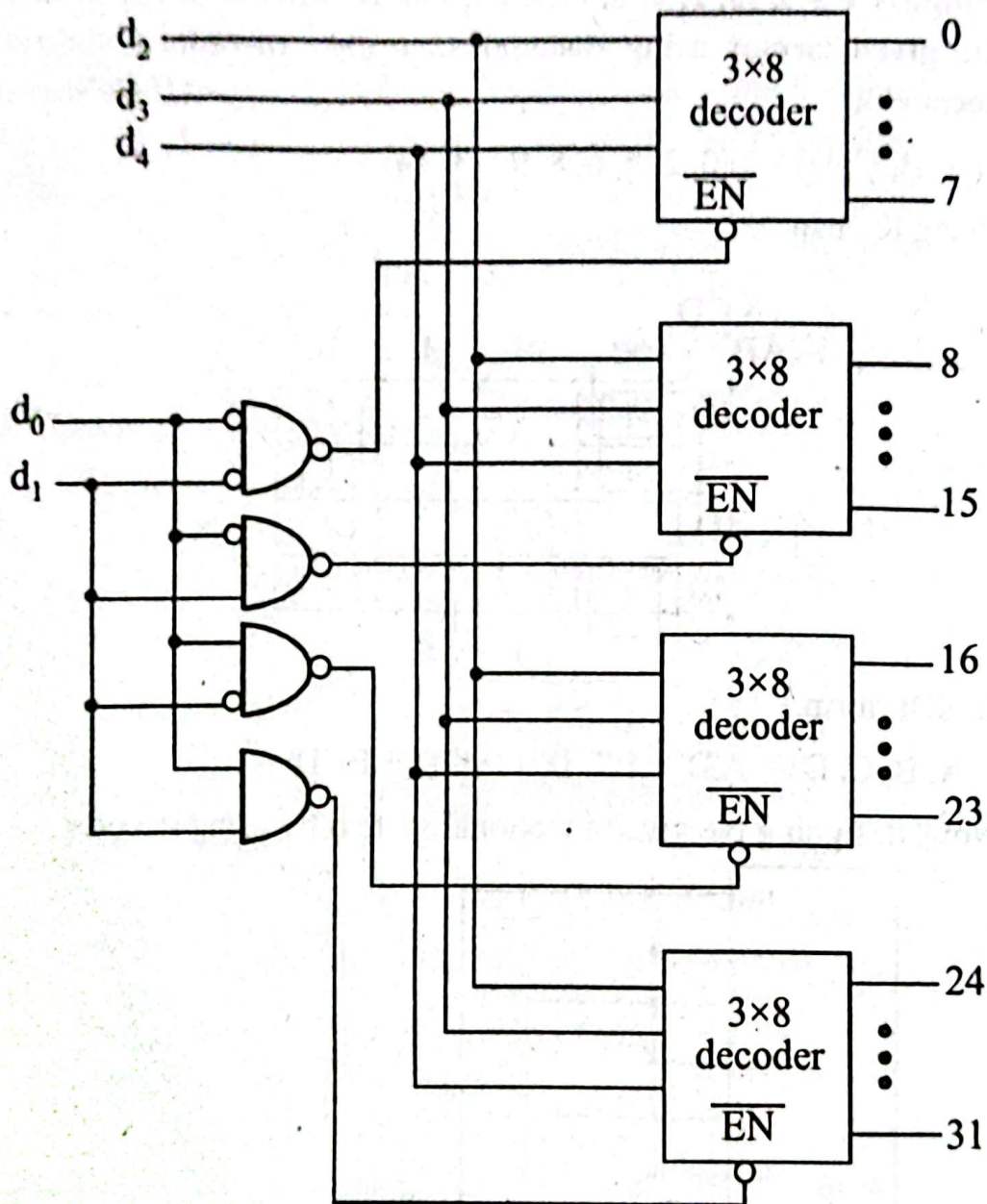


Fig.: Realization of 5×32 decoder using 3×8 decoders and logic gates

6. Determine the parity bit of 100101 following odd parity and write down 1 byte long representation for this number including the parity bit. Why are parity bits important in digital system?

[2065 Shrawan]

⇒

Given message bits: 100101

odd parity bit: 0

8 bit representation: 01001010

Importance of parity bit in digital system:

A Parity bit is a scheme for detecting errors during transmission of binary information. A parity bit is an extra bit included with a binary message to make the number of 1s either odd or even. The message, including the parity bit is transmitted and then checked at the receiving end for errors. An error is detected if the checked parity doesn't to the one transmitted. The circuit that generates the parity bit is called parity generator and the circuit that checks the parity in the receiver is called a parity checker.

7. Write down the truth table for an odd parity generator for any 2-bit word. Design your circuit using any universal gate of your choice. [2065 Shrawan]

Truth table for odd parity generator

Message		Odd parity bit
A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

Using K-map.

		B	
		0	1
A	0	①	0
	1	0	①

$$Y = A'B' + AB = A \odot B$$

Now, implementing using NAND gate,

$$\begin{aligned} Y &= \overline{A} \overline{B} + AB \\ &= \overline{\overline{\overline{A} \overline{B}} + \overline{AB}} \\ &= \overline{\overline{A} \overline{B}} \cdot \overline{AB} \end{aligned}$$

Implementation with AND and OR gates

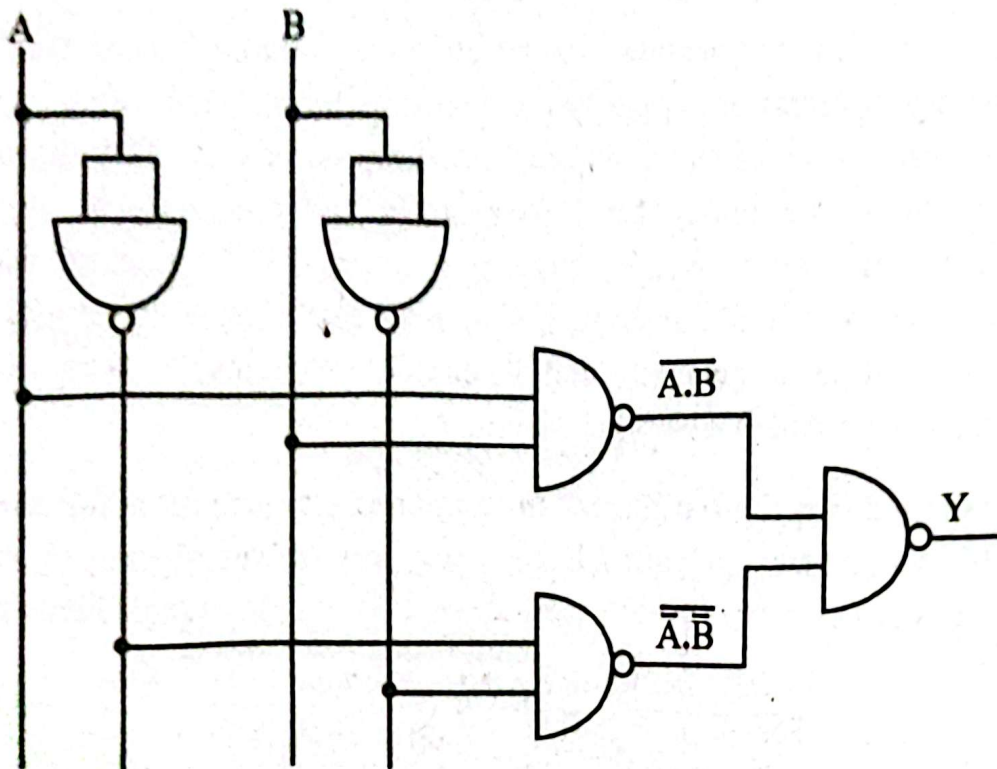


Fig.: X-NOR using NAND

Implement outputs of a full adder circuit by using 8:1 multiplexer.
[2064 Jestha]

⇒ For full adder circuit; let us suppose that A, B and C are inputs and S and C are sum and carry respectively.

A	B	C	S	C
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

$$\text{Sum } [S(x, y, z)] = \Sigma(1, 2, 4, 7)$$

$$\text{Carry } [C(x, y, z)] = \Sigma(3, 5, 6, 7)$$

Now, implementing by using 8:1 multiplexer.

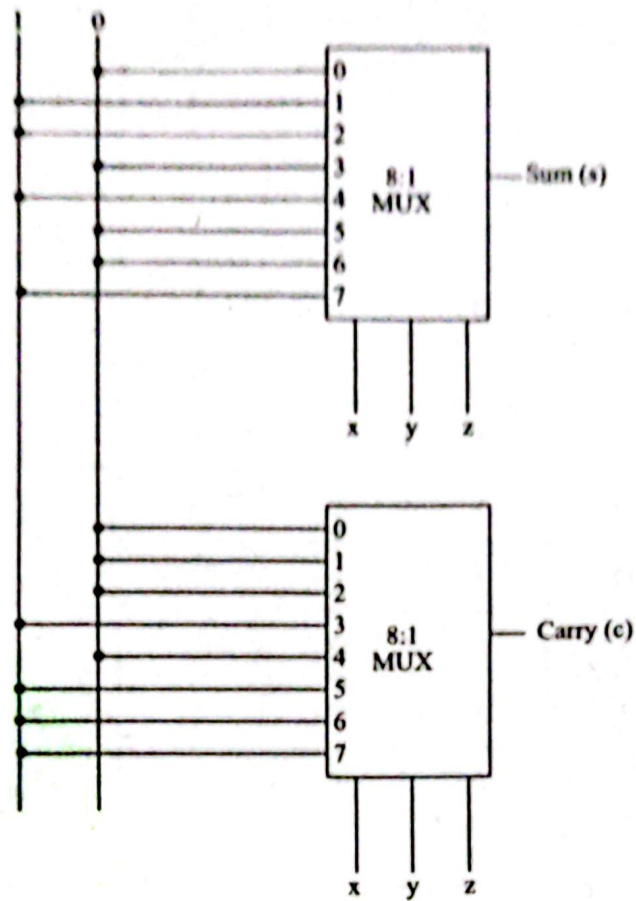
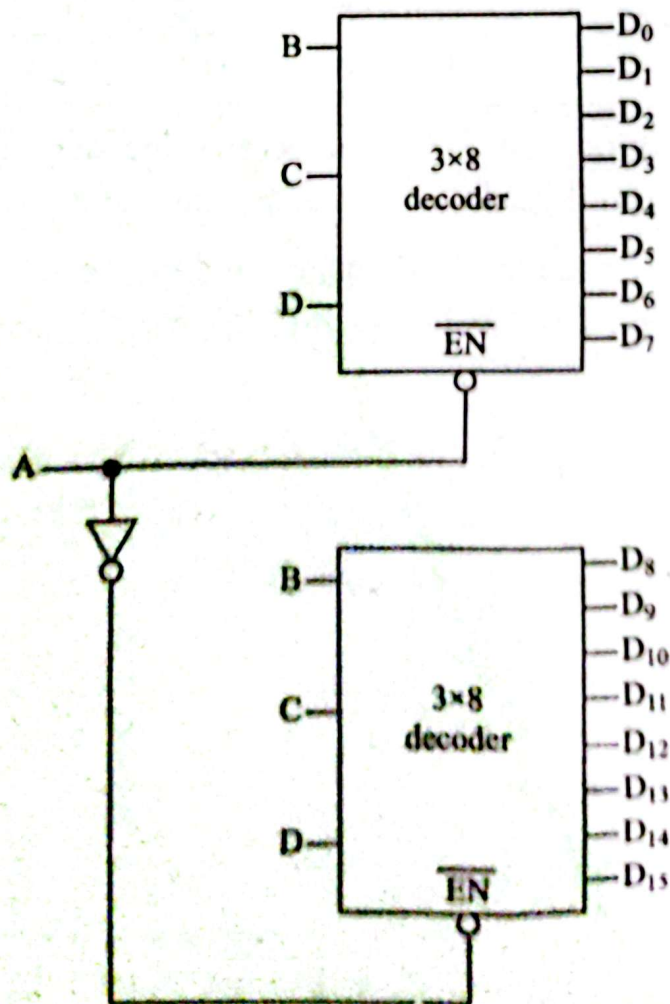


Fig.; Implementation of a full adder using 8:1 multiplexer

9. Construct 4×16 decoder using 3×8 decoder.



A	B	C	D	Y
0	0	0	0	D_0
0	0	0	1	D_1
0	0	1	0	D_2
0	0	1	1	D_3
0	1	0	0	D_4
0	1	0	1	D_5
0	1	1	0	D_6
0	1	1	1	D_7
1	0	0	0	D_8
1	0	0	1	D_9
1	0	1	0	D_{10}
1	0	1	1	D_{11}
1	1	0	0	D_{12}
1	1	0	1	D_{13}
1	1	1	0	D_{14}
1	1	1	1	D_{15}

When $A = 0$, decoder 1 is enabled and decoder 2 is disabled. Hence the output from D_0 to D_7 is high as per the input combination of BCD.

When $A = 1$, decoder 2 is enabled and decoder 1 is disabled. Hence the output from D_8 to D_{15} is high as per the input combination of ABCD.

10. Design a 32:1 MUX using only 8:1 MUX. Use block diagrams.

[2071 Chaitra, 2076 Chaitra]

$\Rightarrow$ We have to design 32: 1 MUX using 8 : 1 MUX.

Given: 8×1 MUX

No. of selection lines is 3.

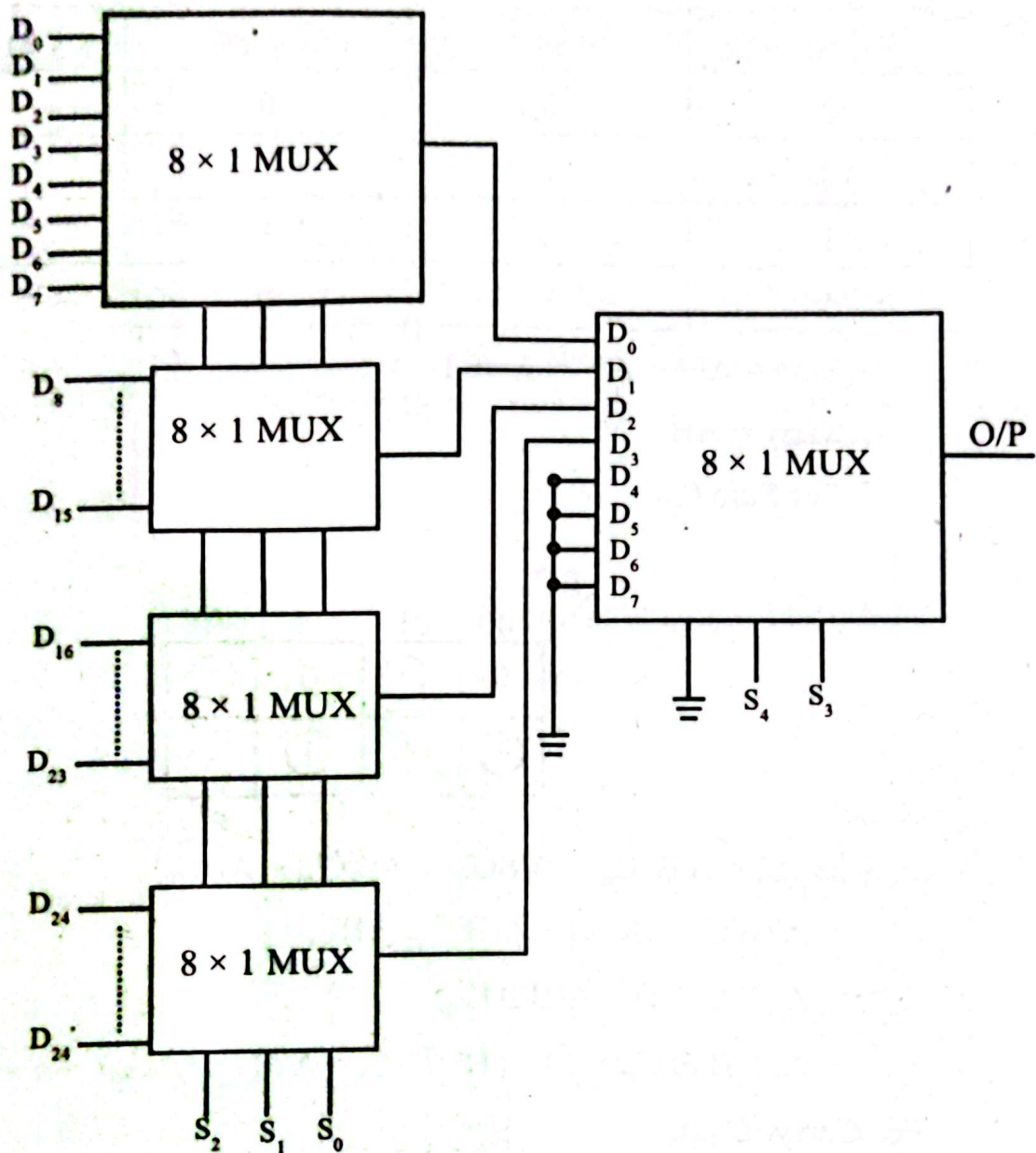
Design: 32×1 MUX

Here, no. of selection lines is 5

Thus to get 32 no. of input lines, we need 4 number of 8×1 MUX.

Let S_4, S_3, S_2, S_1, S_0 be the order of the input lines then S_0, S_1 acts as

selector of MUX and $S_2S_3S_4$ acts as input to the selection line to the input MUX.



11. Show with design that a full-adder can be implemented using two half adders. [2071 Chaitra]

Truth table for full adder

A	B	C_{in}	Sum (S)	Carry (C_{out})
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

Truth table of half adder

A	B	Sum (S)	Carry (C)
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

$$\therefore \text{Sum} = A'B + AB' = A \oplus B$$

$$\text{Carry} = AB$$

For Sum (S),

	BC_{in}			
A	00	01	11	10
0	0	①	0	①
1	①	0	①	0

$$\therefore \text{Sum (S)} = A'B'C_{in} + A'BC'_{in} + AB'C'_{in} + ABC_{in}$$

$$= A'(B'C_{in} + BC'_{in}) + A(B'C'_{in} + BC_{in})$$

$$= A'(B \oplus C_{in}) + A(B \oplus C_{in})'$$

$$= A \oplus B \oplus C_{in} = (A \oplus B) \oplus C_{in} = A \oplus (B \oplus C_{in})$$

For Carry (C_{out}),

	BC_{in}			
A	00	01	11	10
0	0	0	①	0
1	0	①	①	①

$$\therefore \text{Carry (C}_{out}\text{)} = AB + BC_{in} + AC_{in}$$

$$= AB + ABC_{in} + A'BC_{in} + ABC_{in} + AB'C_{in}$$

$$= AB(1 + C_{in}) + A'BC_{in} + AB'C_{in}$$

$$= AB + C_{in}(A \oplus B)$$

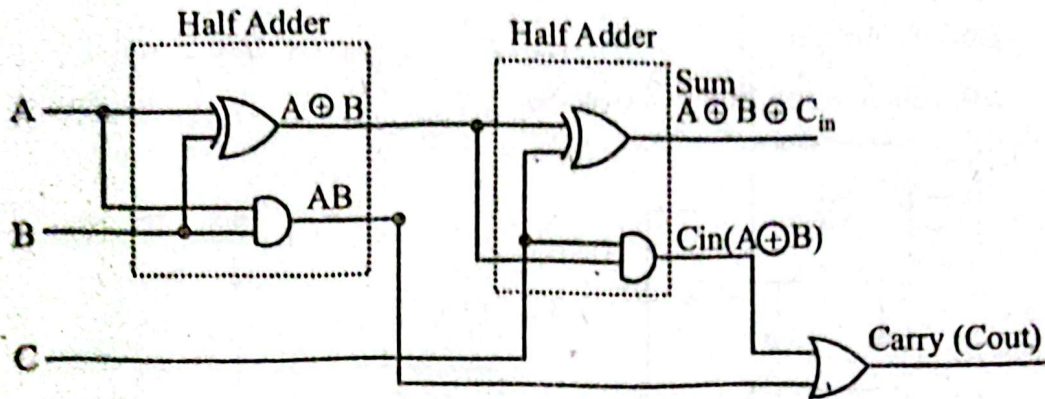


Fig.: Implementation of full adder using two half adders.

12. Design a circuit that compares two 4-bit number, A and B, to check if they are equal. The circuit has one output so that $x = 1$ if $A = B$ and $x = 0$ if $A \neq B$ [2073 Chaitra]

⇒ For the 1-bit numbers to be equal, we have the condition as:

If $A = B$ then $x = A \odot B$

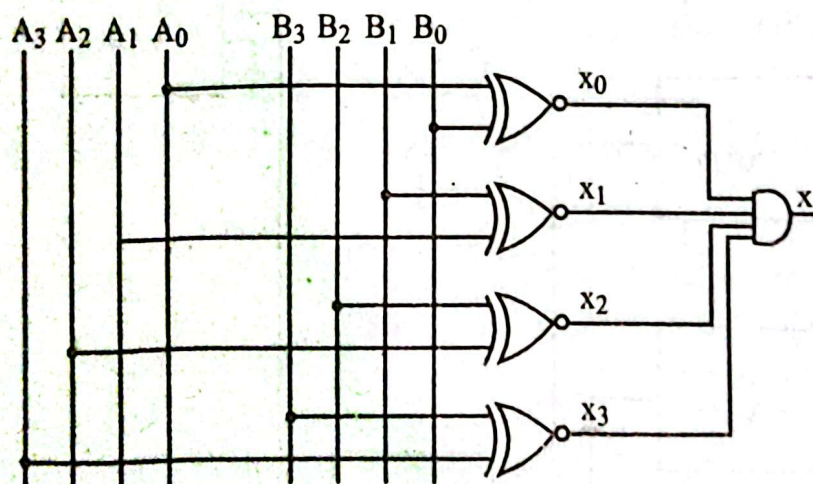
For 4-bit numbers to be equal ($A = B$), all the bits must be equal:

$(A_3 = B_3)$ and $(A_2 = B_2)$ and $(A_1 = B_1)$ and $(A_0 = B_0)$

The above result gives the logic gate formation as:

$(A_3 \odot B_3) \cdot (A_2 \odot B_2) \cdot (A_1 \odot B_1) \cdot (A_0 \odot B_0)$

$= x_3 \cdot x_2 \cdot x_1 \cdot x_0$



13. Design the 32:1 multiplexer using 4:1 multiplexer tree concept and implement the function $F = \sum(0, 1, 3, 8, 9, 13)$ using suitable multiplexer. [2074 Chaitra]

⇒ For 32:1 multiplexer,

no. of inputs = 32 ($I_0, I_1, I_2, \dots, I_{31}$)

no. of outputs = 1

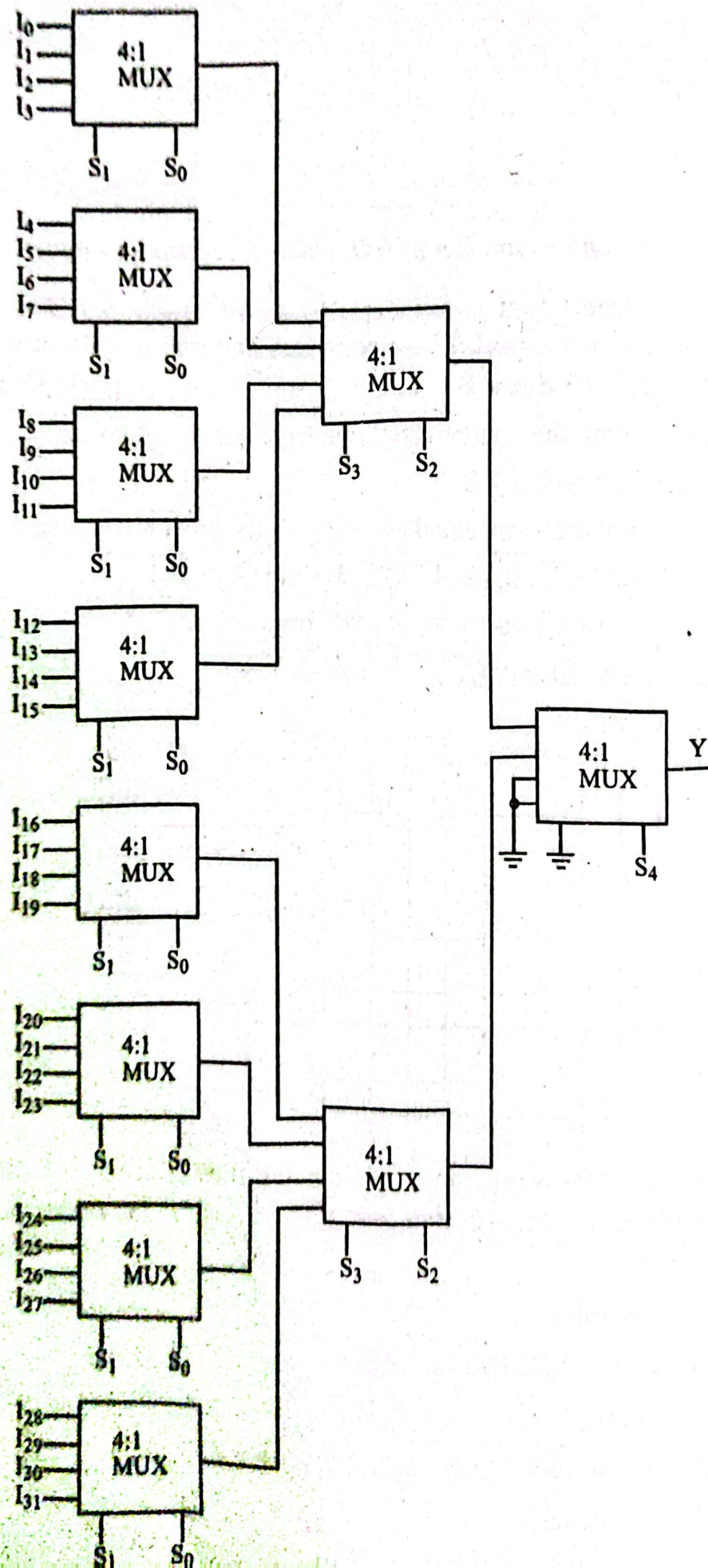
no. of selection lines = 5 (S_4, S_3, S_2, S_1, S_0)

For 4:1 multiplexer,

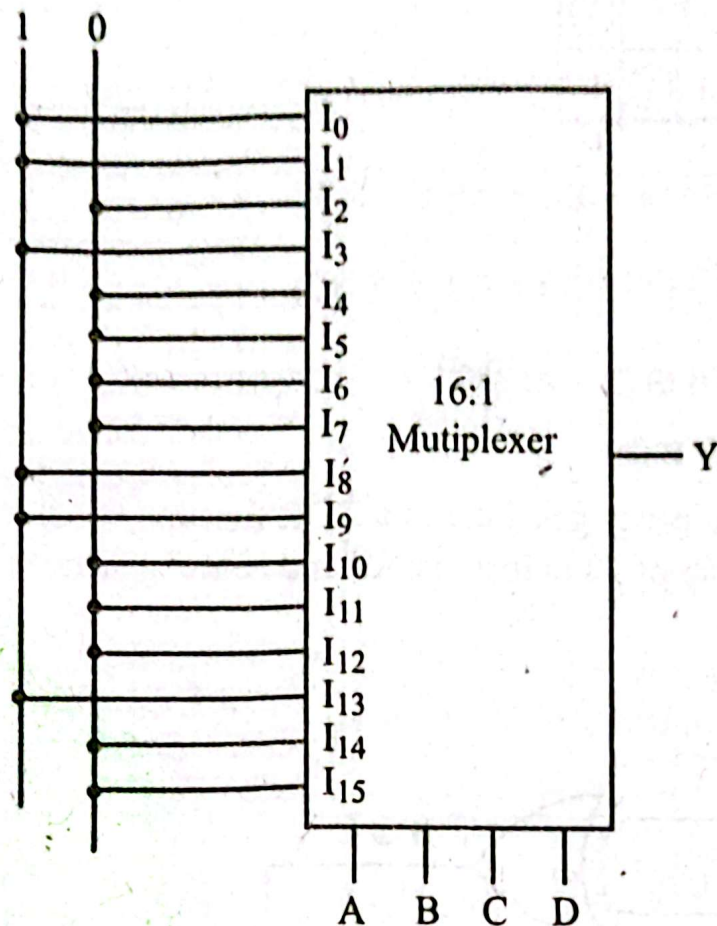
no. of inputs = 4 (I_0, I_1, I_2, I_3)

no. of outputs = 1

no. of selection line = 2 (S_1, S_0)



14. Implement the function $F = \Sigma(0, 1, 3, 8, 9, 13)$ using suitable multiplexer.



15. Explain 3 bits even parity generator circuit clearly [2076 Chaitra]

Solution:

3 bits even parity generator has 3 inputs A, B, C and parity P. P becomes 1 when there are odd number of 1's i.e., message information along with parity should be even for 3 bit even parity generator.

Information			Parity P
A	B	C	
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

Now, Plotting K-map for P,

A \ BC	BC			
	00	01	11	10
0	0	1	0	1
1	1	0	1	0

$$\begin{aligned}
 P &= \bar{A} \bar{B} C + \bar{A} B \bar{C} + A \bar{B} \bar{C} + ABC \\
 &= \bar{A} (\bar{B} C + B \bar{C}) + A (\bar{B} \bar{C} + BC) \\
 &= \bar{A} (B \oplus C) + A (\overline{B \oplus C}) \\
 &= A \oplus B \oplus C
 \end{aligned}$$

Thus, even parity generator is a XOR function which returns 1 when total number of 1's in input is odd and hence, generates an even parity bit.

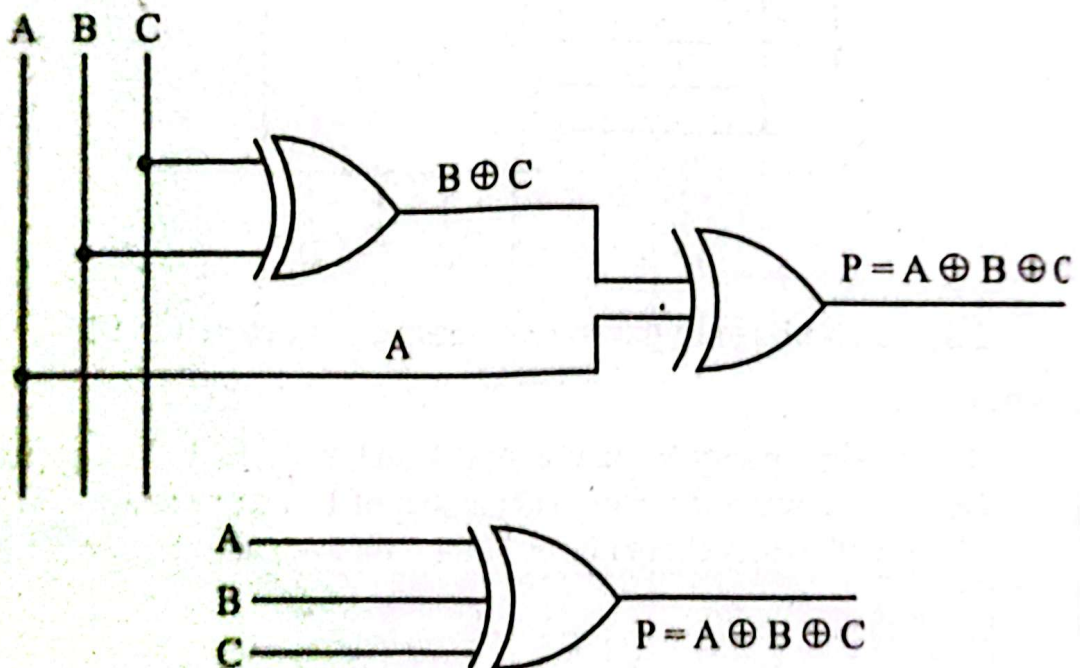


Fig.: Circuit diagram for 3 bit even parity generator

16. Realise a full subtractor circuit by combining only one 1:4 demultiplexer and standard gates [2076 Chitra]

Solution:

A full subtractor circuit has 3 inputs A, B, C and two outputs Difference (D) and Borrow (B_0)

Information			Outputs	
A	B	C	D	B ₀
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

Here, Difference, $D = \Sigma_m(1, 2, 4, 7)$

Borrow, $B_0 = \Sigma_m(1, 2, 3, 7)$

For output D,

$\bar{A}$	0	①	②	3
A	④	5	6	⑦

For output B₀,

$\bar{A}$	0	①	②	③
A	4	5	6	⑦

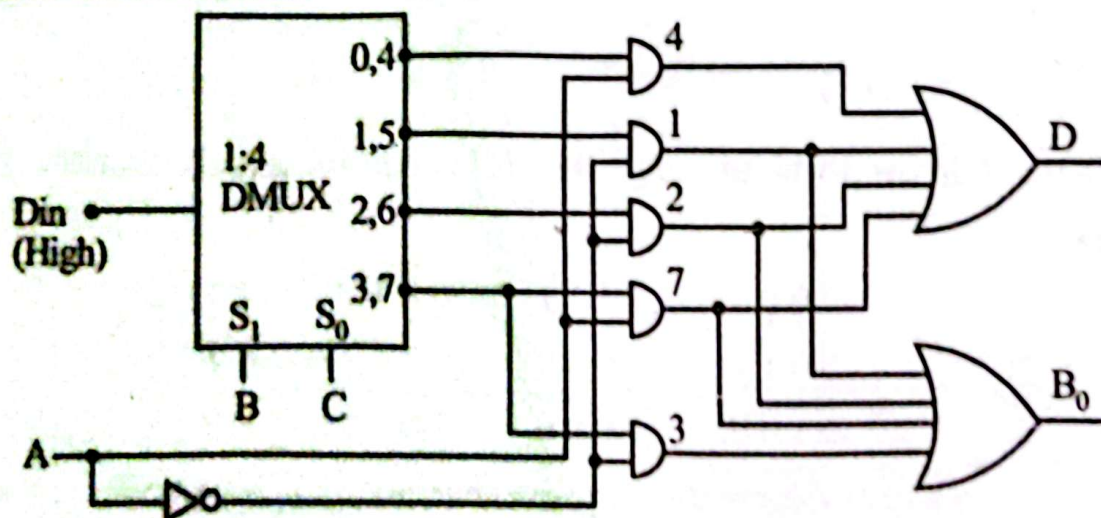


Fig.: Full subtractor using 1:4 demultiplexer and standard gates.